**COLD DESERT**

This graph shows the average annual temperature on the western edge of the Gobi Desert. From November through to March, temperatures fall below freezing.

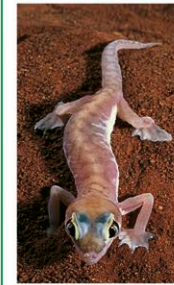
extremely cold. Animals have various ways of coping with this. Most reptiles hibernate, while birds often fly to warmer climates. Mammals keep warm by growing thick fur, or by sheltering underground.

**Explosive breeding**

Desert animals often have highly variable breeding seasons. Instead of reproducing at a fixed time of year, many produce young when there is the best chance of finding food. Female kangaroos, for example, give birth extremely regularly when food is plentiful, but when food is scarce they stop breeding entirely. This flexible system is an efficient way of using resources because it prevents parents having to tend hungry youngsters when they are hungry themselves.

Some desert species carry irregular breeding to extremes. Desert wildlife includes a number of animals that, paradoxically, live or breed in water and, for those species, reproducing is a highly unpredictable and time-sensitive business. Such animals include burrowing frogs and toads, and also freshwater shrimps that live in temporary pools. For months or even years at a time, they are an invisible part of desert wildlife, with the amphibian species lying hidden underground, and the shrimps present only as eggs in dried-up ground. But immediately after a heavy storm, the frogs and toads dig their way to the surface, and the

shrimp eggs hatch. Once active, these animals immediately set about finding mates because they have to complete their life cycles before the pools dry up again.

**MOVING ON SAND**

Flaps of skin between the web-footed gecko's toes make for snowshoelike feet that allow it to run across dunes.

**ATTACK FROM BELOW**

Guided by vibrations overhead, this marsupial mole has emerged from its burrow to ambush a gecko.

**Movement**

Desert sand makes life difficult for animals on the move. Large animals sink into it, while small ones struggle to climb up and down slopes of shifting grains. To combat the problem, some animals, such as golden and marsupial moles, move through the sand rather than above it. Others, such as camels and geckos, have extra large feet, which help to spread their body weight over the surface of the sand and so increase stability. Sidewinding snakes have a different solution: they throw themselves forwards in a succession of sideways jumps, leaving a characteristic pattern of J-shaped tracks. In addition to saving energy, this method helps to minimize contact with hot ground.

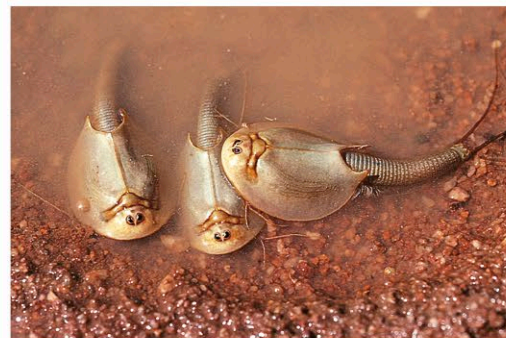
Some insects and lizards have learned to tolerate hot ground by alternating the feet that are in contact with the ground at any one time. Having long legs also helps as they hold the animal's body away from the sand's surface, where the heat is fiercest.

**CHANGING COAT**

In the deserts of central Asia, freezing winters are followed by soaring temperatures in spring. The Bactrian camel manages by growing a thick coat that falls off as soon as the spring warm up begins. This camel is about halfway through this process.

often described as "cold-blooded," but this actually means that their body temperature rises and falls with that of their surroundings. Although they thrive in warm habitats, and can survive with a body temperature of up to 111°F (44°C), they often have to sit out the hottest part of the day in shade. Some desert birds cool down by panting, which involves fluttering the flap of skin over their throats. Desert kangaroos and wallabies lick their front legs, covering them with saliva. As the saliva evaporates, the animal's blood cools down.

In high-latitude desert regions, such as the Gobi Desert of central Asia and the Great Basin Desert of North America, winter can be

**LIFE IN BRIEF**

Trapped in a rapidly shrinking desert pool, these adult tadpole shrimps have only a few days to live, but the eggs they leave behind can survive in a dried-out state for several years—long enough to last until the next heavy storm, when they will hatch.

**NOCTURNAL ACTIVITY**

Desert jerboas are typical of the small rodents that live in dry habitats. They are entirely nocturnal, and feed mainly on seeds. Jerboas can travel over 6 miles (10 km) in their search for food, hopping on their long back legs and balancing with their tails.

**NOMADIC ANIMALS**

Where food and water supply is patchy, some animals adopt a wandering lifestyle. This is common in desert habitats, especially in animals that can fly. Desert locusts are famous for their huge nomadic swarms, and some desert birds, particularly seed eaters, form large nomadic flocks. Unlike migrants, nomadic animals do not follow fixed routes—the weather often dictates their course—and they breed erratically, wherever they find a good food supply.

**WILD BUDGERIGARS**

Australia's nomadic birds include the budgerigar, the cockatiel, and several pigeon species. Budgerigars can breed when just one month old; and, since parents can raise several families in quick succession, flocks can build to prodigious numbers.

